

Part A — Simple Class and Object

```
<?php
class Student
{
    function sayHello()
    {
        echo "Hello! I am a student.";
    }
}
$student1 = new Student();
$student1->sayHello();
?>
```

Output:

Hello! I am a student.

Part B — Class with Constructor

```
<?php
class Student
{
    public $name;
    public $studentId;
    public $department;
    function __construct($name, $studentId, $department)
    {
        $this->name = $name;
        $this->studentId = $studentId;
        $this->department = $department;
    }
    function showInfo()
    {
        echo "Name: " . $this->name . "<br>";
        echo "Student ID: " . $this->studentId . "<br>";
        echo "Department: " . $this->department . "<br>";
    }
}
$student1 = new Student(
    "Ahmad",
    1001,
```

```
        "Computer Science"
    );
    $student1->showInfo();
?>
```

Output:

Name: Ahmad

Student ID: 1001

Department: Computer Science

Part C — Create Another Object

```
$student2 = new Student(
    "Sara",
    1002,
    "Information Systems"
);
$student2->showInfo();
```

Output:

Name: Sara

Student ID: 1002

Department: Information Systems

Questions:

1. How many classes did you create?
I have created one single class called Student that has five class members, three properties and two methods.
 2. How many objects did you create?
I have created 2 objects till now from the Student class, one called student1 and another one student2
-

Part D — Access Modifiers

```
<?php
class BankAccount
{
    public $ownerName;
    private $balance;
```

```

function __construct($ownerName, $balance)
{
    $this->ownerName = $ownerName;
    $this->balance = $balance;
}
function showBalance()
{
    echo "Balance: " . $this->balance . "<br>";
}
}
$account1 = new BankAccount(
    "Ahmad",
    5000
);
echo "Owner: " . $account1->ownerName . "<br>";
$account1->showBalance();
?>

```

Output:

Owner: Ahmad

Balance: 5000

Questions:

Try this: `echo $account1->balance;`

No, it will not work, Because `$balance` is declared as private. A private property can only be accessed inside the class where it is declared. A public property can be accessed from outside the class.

`echo $account1->ownerName;`

It will work because the `ownerName` is a public property.

Part E — Simple Inheritance

```

<?php
class Person
{
    public $name;
    function __construct($name)
    {

```

```

        $this->name = $name;
    }
    function introduce()
    {
        echo "My name is " . $this->name . "<br>";
    }
}
class Student extends Person
{
    function study()
    {
        echo $this->name . " is studying.";
    }
}
$student1 = new Student("Ahmad");
$student1->introduce();
$student1->study();
?>

```

Output:

My name is Ahmad

Ahmad is studying.

Part F — Understanding Inheritance

Question 1: Which class the parent class?

Answer 1: Person() is the parent class

Question 2: Which class is the child class?

Answer 2: Student()

Question 3: Which keyword creates the inheritance relationship?

Answer 3: 'extends' is the keyword that makes a class child of another class.

Question 4: The introduce() method was written inside which class?

Answer 4: Inside Person() class.

Question 5: Can the \$student1 object call introduce()?

Answer 5: Yes, because it inherits it from Person().

Question 6: Why can \$student1 use introduce() even though it is not written inside the Student class?

Answer 5: Because Student inherits from Person using extends. Therefore, the child class Student can use the methods and properties inherited from its parent class Person.

Part G — Vehicle and Car

```
<?php class Vehicle
{
    protected $brand;

    function __construct($brand)
    {
        $this->brand = $brand;
    }

    function start()
    {
        echo "The vehicle is starting.<br>";
    }
}
class Car extends Vehicle
{
    function showBrand()
    {
        echo "Car brand: " . $this->brand;
    }
}

$car1 = new Car("Toyota");
$car1->start();
$car1->showBrand();
?>
```

Output:

The vehicle is starting.

Car brand: Toyota

